

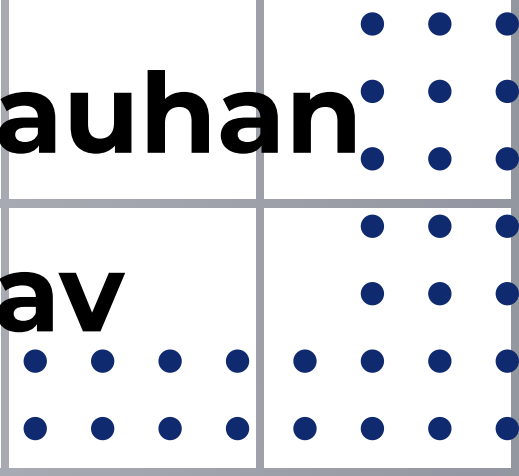


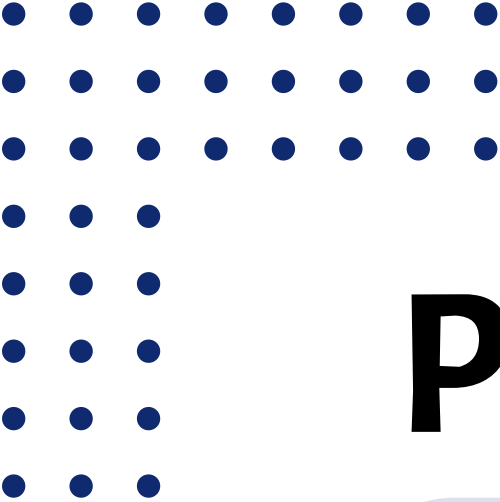
SMART STORAGE OPTIMIZER

**A Usage-Based Intelligent Storage
Management System.**

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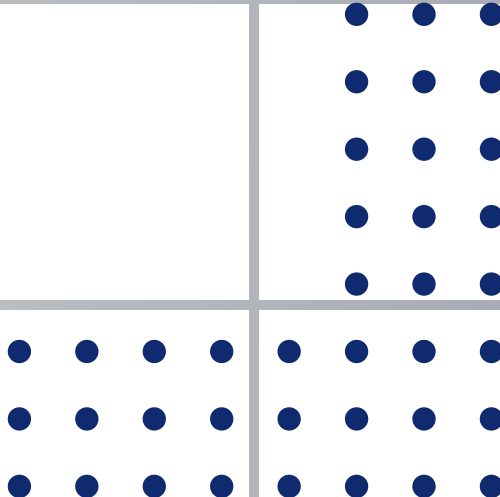
PROBLEM-

Modern smartphone users frequently face storage issues due to increasing digital content.

Main problems:

- **Large photos and videos consume significant storage**
- **Duplicate files are stored unknowingly**
- **Old and unused files occupy unnecessary space**
- **Temporary/cache files accumulate over time**

Manual deletion is time-consuming and inefficient.



MOTIVATION

Current solutions are not enough for students and everyday users.

Why this project?

- **Buying more storage is expensive**
- **Upgrading devices is not always affordable**
- **Cloud storage needs better management**
- **Users need a smart and automated solution**

Goal: Efficiently optimize existing storage without extra cost.



EXISTING SOLUTIONS

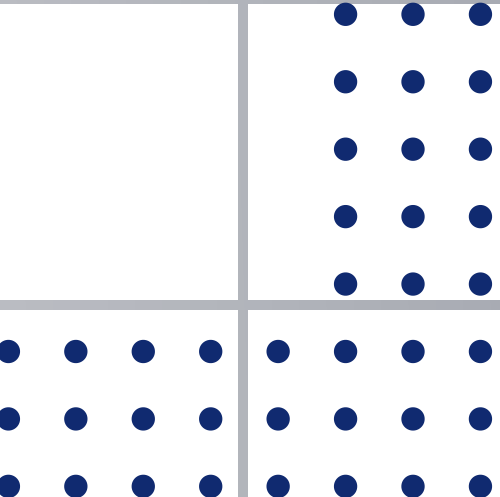
Existing solutions:

- **Built-in phone storage cleaners**
- **Cache cleaners**
- **Cloud storage platforms**

Limitations:

- **Only clear temporary files**
- **Cannot intelligently detect duplicates**
- **No usage-based file recommendations**
- **No smart cloud allocation**

Gap: Lack of integrated intelligent storage optimization.



PROPOSED SOLUTION

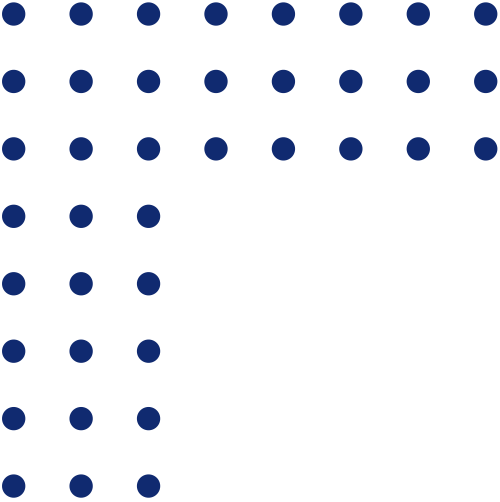
Smart Storage Optimizer

An intelligent storage management system that:

- Scans local folders / uploaded files
- Extracts file metadata
- Detects duplicate files using hash comparison
- Identifies unused and large files

Benefits:

Saves storage space
Reduces manual effort
Improves device performance

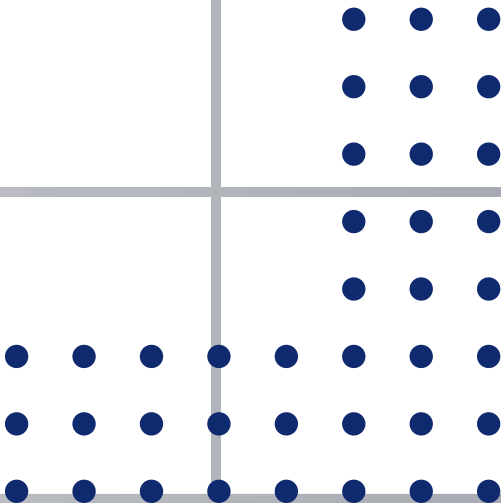


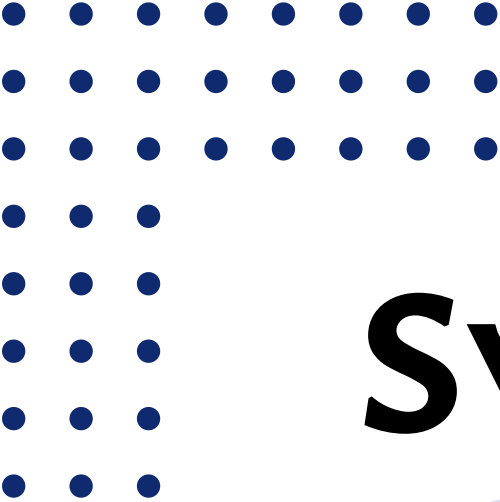
TECHNOLOGY

Software Used:

- Python (core logic)
- Streamlit (user interface)
- Pandas (data handling)
- Hashlib (duplicate detection)
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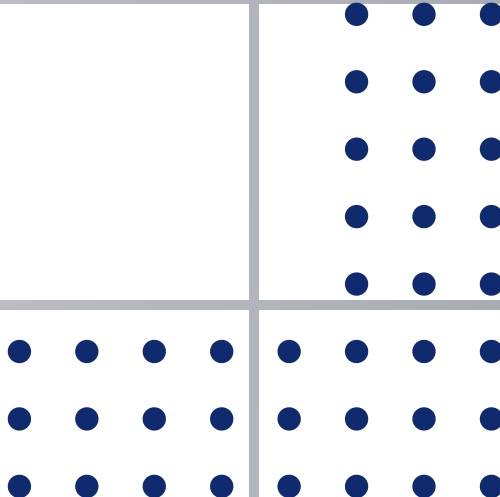
Platform:

- Desktop prototype (simulating Android behavior)
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SYSTEM ARCHITECTURE

Workflow:

1. **User selects folder or uploads files**
 2. **System scans all files**
 3. **Metadata collected:**
 - **File size**
 - **Last modified time**
 - **File type**
 4. **Hash generated for duplicate detection**
 5. **Files scored based on priority**
 6. **Recommendation provided**
- Output: File optimization decision dashboard**
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How duplicates are identified:

- **Hash values generated for each file**

- **Same hash = duplicate file**

Advantages:

- **Accurate file comparison**

- **Faster than manual checking**

- **Reduces redundant storage**

Example: Duplicate images/videos can be safely removed.

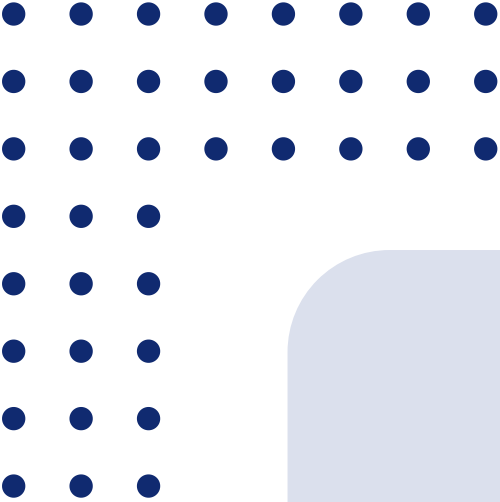
File Scoring Based On:

- **File size**
- **File age**
- **Duplicate status**
- **Temporary file extension**

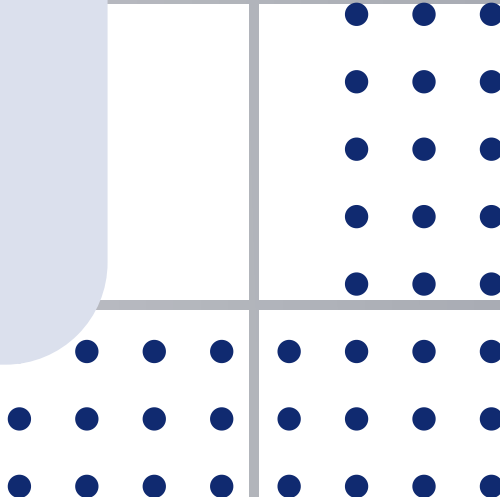
Routing Decisions:

- **Duplicate → Delete**
- **Sensitive files → USB Backup**
- **High score → Permanent Cloud**
- **Medium score → Temporary Cloud**
- **Low score → Keep Local**

This makes the system intelligent and automated



Features implemented in prototype:

- **Folder scanning on local system**
 - **File upload option**
 - **Duplicate file detection**
 - **Large file identification**
 - **Storage summary dashboard**
 - Dashboard shows:**
 - **Total files scanned**
 - **Duplicate files count**
 - **Total storage occupied**
 - **Recommended actions**
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RESULT AND OUTPUT

- **Prototype Results:**
- **Successfully scanned files and folders**
- **Detected duplicate files accurately**
- **Highlighted large storage-consuming files**
- **Suggested optimized storage decisions**

- **Impact:**
- **Better storage utilization**
- **Reduced unnecessary storage usage**
- **User-friendly recommendations**

<https://pbl-project-c3yfgtfrktevavbbqfrqey.streamlit.app/>



StorageMind

AI Inspired Storage Optimization Engine

Smart file analysis using hashing, storage scoring, and intelligent routing.

Choose Input Method

Upload files to scan

 Upload 200MB per file

 Scan Demo Folder

 Scan Uploaded Files



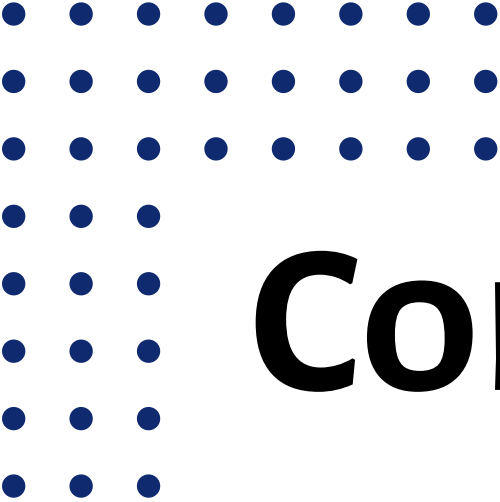
File Decisions

	name	path	size_kb	days_old	ext	is_dupe
2	images (3).jpeg	,vscode2/demo2/images (3).jpeg	7.810000	0	.jpeg	☐
3	download.jpeg	,vscode2/demo2/download.jpeg	5.190000	0	.jpeg	☐
4	download (5).jpeg	,vscode2/demo2/download (5).jpeg	7.200000	0	.jpeg	☐
5	images (6).jpeg	,vscode2/demo2/images (6).jpeg	7.100000	0	.jpeg	☒
6	images (8).jpeg	,vscode2/demo2/images (8).jpeg	8.280000	0	.jpeg	☐
7	download (1).jpeg	,vscode2/demo2/download (1).jpeg	6.040000	0	.jpeg	☐
8	images (11).jpeg	,vscode2/demo2/images (11).jpeg	6.120000	0	.jpeg	☐
9	download (3).jpeg	,vscode2/demo2/download (3).jpeg	7.200000	0	.jpeg	☒
10	images (1).jpeg	,vscode2/demo2/images (1).jpeg	9.720000	0	.jpeg	☐
11	download (2).jpeg	,vscode2/demo2/download (2).jpeg	18.190000	0	.jpeg	☐

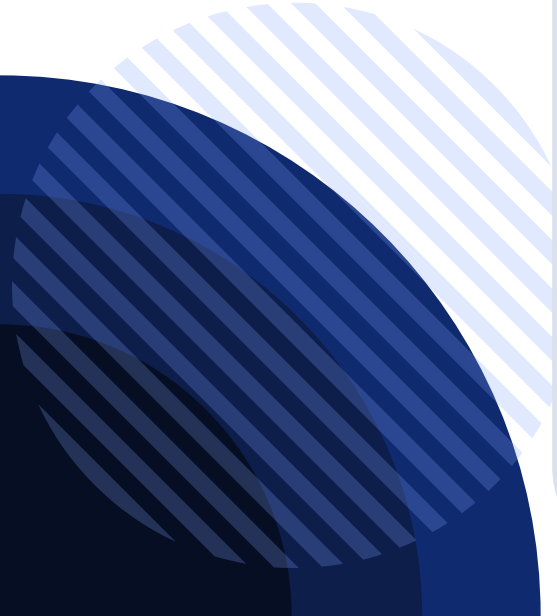
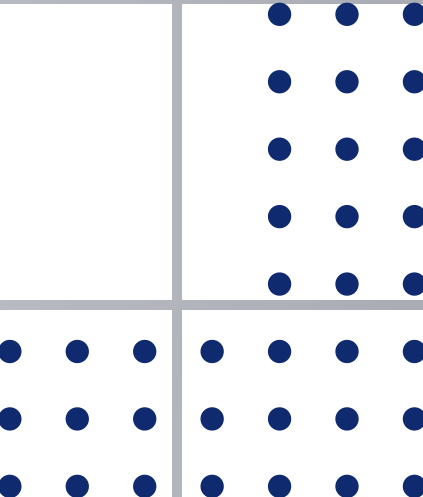
FUTURE SCOPE

The Smart Storage Optimizer can be further improved in the following ways:

- **Develop a full Android application for real-time mobile storage management**
- **Integrate with cloud platforms like Google Drive and Dropbox for automatic backup**
- **Add AI-based usage prediction to identify files that are rarely accessed**
- **Introduce scheduled auto-cleanup for temporary and junk files**
- **Enable automatic transfer of less frequently used files to secondary storage (such as SD cards or external drives), reducing the need for manual file movement and saving user time**



CONCLUSIONS

- **The Smart Storage Optimizer provides a practical and cost-effective solution to storage management problems.**
 - **It helps users:**
 - **Save storage space**
 - **Detect duplicates**
 - **Reduce clutter**
 - **Improve device efficiency**
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THANK YOU

